Experiment-1

Name of the experiment: Preparation of rock samples for testing

1. SCOPE

In order to obtain valid results from tests on rock materials, careful and precise specimen preparation is required. This method outlines preparatory procedures recommended for normal rock mechanics tests.

2. COLLECTION AND STORAGE

Test material is normally collected from the field in the form of drilled cores. Field sampling procedures should be rational and systematic, and the material should be marked to indicate its original position and orientation relative to identifiable boundaries of the parent rock mass.
Good practice generally dictates that laboratory tests be made upon specimens representative of field conditions. Thus, it follows that the field moisture condition of the specimen should be preserved until the time of the test.
Ideally, samples should be moisture proofed immediately after collection either by waxing, spraying, or packing in polyethylene bags or sheet. (Example: For moisture proofing by waxing, the following procedure can be used for core that will not fail apart in handling.

a) Wrap core in a clear thin polyethylene,
b) Wrap in cheese cloth,
c) Coat wrapped core with a lukewarm wax mixture to an approximate6 mm thickness. The wax should consist of a l to 1 mixture of paraffin and microcrystalline wax.

They should be transported as a fragile material and protected from excessive changes in humidity and temperature. The identification markings of all samples should be verified immediately upon their receipt at the laboratory, and an inventory of the samples received should be maintained.
Samples should be examined and tested as soon as possible after receipt; however, it is often necessary to store samples for several days or even weeksto complete a large testing program. Every care must be taken to protect stored samples against damage. Core logs of samples should be available.

3. AVOIDANCE OF CONTAMINATION

The deformation and fracture properties of rock may be influenced by air, water, and other fluids in contact with their internal (crack and pore) surfaces. If these internal surfaces are contaminated by oils or other substances, their properties may be altered appreciably and give misleading test results. Of course, a cutting fluid is required with many types of specimen preparation equipment. Clean water is the preferred fluid. Even so, one must be cognizant of the effect of moisture on the test specimens. While it may be impossible to exactly duplicate the in situ conditions even if they were known, a concerted effort should be made to simulate the environment from which the samples came. Generally, there are three conditions to be considered:

a) Hard, dense rock and low porosity will not normally be affected by moisture. This type of material is normally allowed to air-dry prior to testing to bring all samples to an equilibrium condition. Drying at temperatures above 120°F (49°C) is not recommended as excessive heat may cause an irreversible change in rock properties.
b) Some shales and rocks containing clay will disintegrate if allowed to dry. Usually the disintegration of diamond drill cores can be prevented by wrapping the cores as they

are drilled in a moisture proof material such as aluminum foil or chlorinated rubber, or sealing them in moisture proof containers.

c) Mud shales and rock containing bentonites (e.g., tuff) may soften if the moisture content is too high. Most of the softer rocks can be cored or cut using compressed air to clear cuttings and to cool the bit or saw.

It is imperative to determine very early in the test program the moisture sensitivity of all types of material to be tested and to take steps to accommodate the requirements throughout the test life of the selected specimens.

4. SAMPLE SELECTION

The judicious sample selection is important for overall success of the testing programme. It follows, therefore, that test specimens should be of good quality unweathered material, should be free from incipient fractures or damage and should be of regular shape and size.

The timing of sample collection may also play an important role with regard to the end result of tests, in that collection of samples should coincide with the availability of the facilities for specimen preparation, thus avoiding long delays and storage problems which, in turn, may result in deterioration and unnecessary additional costs.

Closest teamwork of the laboratory personnel and the project engineer/geologist must be continued throughout the testing program since, as quantitative data become available, changes in the initial allocation of samples or the securing of additional samples may be necessary.

5. CORING

Virtually all laboratory coring is done with thin-wall diamond rotary bits, which may be detachable or integral to the core barrel. The usual size range for laboratory core drills is from 6-in.-diam (152.4-mm) down to l-in.(25.4-mm) outside diameter. Typical sample diameters for uniaxial testing are2.125 in. (54 mm). Drilling machines range from small quarry drills to modified machine shop drill presses. Almost any kind can be adapted for rock work by fitting a water swivel, but a heavy, rigid machine is desirable in order to assure consistent production of high quality core. The work block must be clamped tightly to a strong base or table so as to prevent any tilting, oscillation, or other shifting. To avoid unnecessary unclamping and rearrangement of the work block, it is desirable to have provision for traversing the drill head or the work block. Traversing devices must lock securely to eliminate any play between drill and work. The drill travel should be sufficient to permit continuous runs of at least 10 to 12 in. (254 to 304.8 mm), without need for stopping the machine. Optimum drilling speeds vary with bit size and rock type, and to some extent with condition of the bit and the characteristics of the machine. The general trend is that drill speed increases as drill diameter decreases; also, higher drill speeds are sometimes used on softer rocks. The broad range of drill speeds lies mainly between 200 and 2,000 rpm. No hard-and-fast rules can be given, but an experienced operator can easily choose a suitable speed by trial. Some core drills are hand-fed, but it is desirable to have some provision for automatic feed. The ideal feed arrangement is a constant-force hydraulic feed which can be set for each bit size and rock type, but such machines are quite rare. Constant-force feed can be improvised by means of a weight and pulley arrangement. On adapted metalworking drill presses, the automatic feed rate for a given drill size and rock type can be determined; however, since there is a danger of damaging the machine or the core barrel if too high a feed rate is used, an electrical overload breaker should be provided.

6. SAWING

For heavy sawing, a slabbing saw is adequate for most purposes. For exact sawing, a precision cutoff machine, with a diamond abrasive wheel about10 in. (254 mm) in diameter, and a table with two-way screw traversing and provision for rotation are recommended. The speed of the wheel is usually fixed, but the feed rate of the wheel through the work can be controlled.
Clean water, either direct from house supply or re-circulated through a settling tank, is the standard cutting and cooling fluid. For crosscutting, core should be clamped in a vee-block slotted to permit passage of the wheel. By supporting the core on both sides of the cut, the problem of spalling and lip formation at the end of the cut is largely avoided. Saw cuts should be relatively smooth and perpendicular to the core axis in order to minimize the grinding or lapping needed to produce end conditions required for the various tests.

Fig.1: Rock coring and cutting machines for preparation of rock samples

7. END PREPARATION

Due to the rather large degree of flatness required on bearing surfaces for many tests, end grinding or lapping is required. Conventional surface lapping provided the most practical means of preparing flat surfaces. Procedures are essentially comparable to metal working. Quite often a special jig is constructed to hold one or more specimens in the lapping operation. For core diameters of 2-1/8 in. (54 mm) or less, a lap can be used for grinding flat end surfaces on specimens, although producing a sufficiently flat surface by this method is an art. To end-grind on the lap, a cylindrical specimen is placed in a steel carrying tube which is machined to accept core with a clearance of about0.002 in. (0.0508 mm). At the lower end of this tube is a steel collar which rests on the lapping wheel. The method requires use of grinding compounds and, hence, is not recommended where other methods are available.

8. SPECIMEN CHECK

After the specimen has been finally ground to size, it should be submitted for inspection before testing. The checking should take account of dimensional accuracy and the conditional integrity of the specimen relative to the test concepts. In general terms, test specimens should be straight, their diameter should be constant, and the ends should be flat, parallel, and normal to the long axis. Final dimensions of the specimen are normally measured with a dial gauge, vernier caliper, Vee block etc. and reported as listed below (sample for uniaxial compressive strength according to ISRM-1978).

a) The diameter of the specimen should be measured to the nearest 0.1mm and its length-to-diameter ratio should be checked to have between 2.5 to 3.0.
b) The ends of the specimen should be ground or lapped flat to within 0.02 mm.
c) The ends of the specimen should be square to the perpendicular axis and to be within 0.001radian or 0.05mm in 50mm.
d) The sides of the specimen should be smooth and free from any irregularities, and be straight to within 0.3mm over the entire length.
e) Specimens should not be tested which do not meet the dimensional tolerances specified in the respective test methods.

9. REQUIREMENT OF MINIMUM NUMBER OF SPECIMENS

Test results obtained by testing individual specimens sometimes indicate a high degree of scatter and not always, representative of the present rock body. Therefore, it often necessary to conduct a greater number of test samples in order to limit costs without compromising accuracy or reliability. There are various statistical methods for obtaining minimum number of samples required for an individual test on a particular rock type. However, one of the methods considering varying degree of confidence is given below.

$$\text{Standard deviation} = S = \sqrt{\frac{\sum (x_i - x_m)^2}{N-1}}$$

V= coefficient of variation in the preliminary test of N specimen $= \dfrac{S}{x_m} \times 100$

The following relationship can be used for determination of number of specimens 'n' to be tested

$$n = \left[\frac{(P+1)tV}{100(P-1)} \right]^2$$

where,

P = a number depends on permissible deviation of mean value $= \dfrac{100+r}{100-r}$

r = permissible deviation of mean value or error

(i.e. forr =10% P = 1.22)

t= depends on the confidence desired and the number of specimens tested (N) in a preliminary experimental test (See table). (i.e. for N = 4 t = 3.18 for 95 % confidence level)

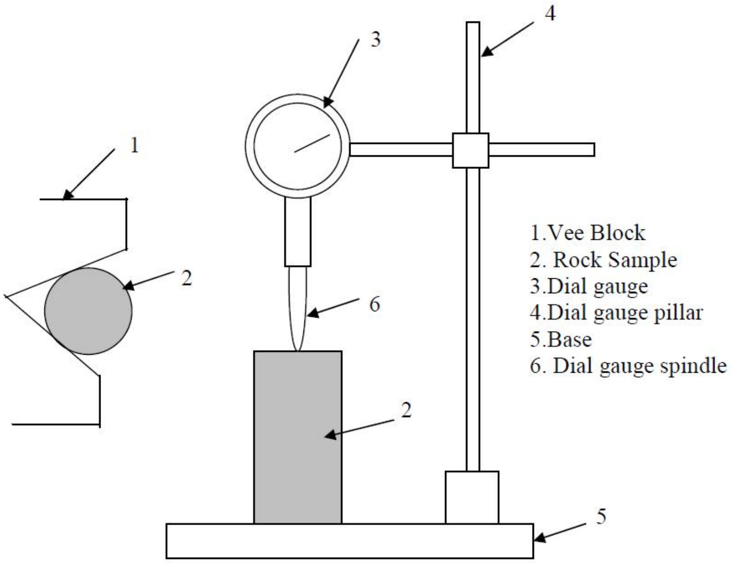

Fig.1: Typical rock sample inspection setup.

Table: Some critical value of ‘t’

N	DF*	t: 90	95	99 Confidence level %
3	2	2.92	4.30	9.92
4	3	2.35	3.18	5.84
5	4	2.13	2.78	4.60
6	5	2.02	2.57	4.03
7	6	1.94	2.45	3.71
8	7	1.90	2.36	3.50
9	8	1.86	2.31	3.36
10	9	1.83	2.26	3.25
11	10	1.81	2.23	3.17
12	11	1.80	2.20	3.11

* Degrees of freedom

References:

International Society for Rock Mechanics Suggested Methods "Rock Characterization, Testing and Monitoring" Editor E.T. Brown, Pergamon Press 1981.